A Fly Visual-to-Motor Circuit Plays Snake: a Minimal Connectome-Constrained Controller

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Abstract

The FlyWire reconstruction of an adult *Drosophila* brain makes it possible to use measured wiring, rather than an arbitrary architecture, as the recurrent core of an artificial agent. In this study, 133 neurons lying on short paths from visual projection neurons to descending neurons are extracted, their 650 connections are converted into fixed weights from synapse counts and predicted neurotransmitters, and a trainable interface of 157 parameters is attached. Only that interface is optimised, by an evolution strategy, in roughly one minute per run on a laptop CPU, while the connectome weights stay frozen. The resulting agent plays Snake on a 10×10 grid: 27.3 ± 3.6 food items are eaten per game over 500 unseen games and 10 independent runs, and no game ends at a wall. Degree-preserving shuffles, Erdős–Rényi graphs, sign-shuffled weights and a network without recurrent connections reach the same score range, so no measurable benefit of the measured wiring is found on this task. The agent, the controls and the main failure mode (self-collisions) are reported as a small, fully reproducible case study rather than as evidence about fly computation.

Keywords: connectome; *Drosophila*; FlyWire; recurrent network; evolution strategy; game agent

1 Introduction

Electron-microscopy reconstructions now provide synapse-level wiring diagrams of entire insect brains. The FlyWire dataset describes 5×10^7 chemical synapses between 139,255 neurons of an adult female *Drosophila melanogaster* [1]. It is accompanied by whole-brain annotations of cell classes and types [7] and by neurotransmitter identities predicted from the electron-microscopy images [2]. Such data invite a simple question: if the wiring is fixed and only the coupling to the outside world is learned, what behaviour can a measured circuit support?

Related work has mostly been aimed at explaining neural activity or function. A model of the fly motion pathways built on measured connectivity, with unknown single-neuron and synapse parameters optimised for motion detection, was shown to predict recorded neural responses [5]. Control-theoretic analysis of the *C. elegans* connectome was used to predict which neurons are involved in locomotor behaviour [10]. In the present study a different angle is taken: a connectome fragment is placed in a closed loop with a game. Snake is convenient because the rules are short, the state is small, and strong reference strategies are easy to write by hand. Games of this kind are usually approached with deep reinforcement learning, as in the Atari benchmark [6] and in Snake itself [8]. Maximum performance is not the aim here; the learned part is kept as small as possible so that whatever the agent does rests mainly on the fixed circuit.

The setup is close in spirit to reservoir computing, in which a fixed random recurrent network is combined with a trained readout [4]. The difference is that the reservoir used here is not random: it is a measured piece of the fly brain. It starts at visual projection neurons, which convey information on the presence and location of visual features to the central brain and whose activation can evoke avoidance-like and turning behaviours [9], and it ends at descending neurons. Matched random and shuffled reservoirs are therefore also trained as controls.

Three contributions are made: (i) a laptop-scale, fully seeded pipeline from FlyWire tables to a closed-loop Snake agent; (ii) a comparison of the measured circuit with four matched alternatives under an identical training budget; and (iii) a negative result on the role of the wiring, released with code, logged results and a browser demo that reproduces the reported games exactly.

2 Methods

Circuit. Connection rows of the FlyWire v783 Codex export were summed per neuron pair, which gave 3,732,460 connected pairs among 139,255 neurons. Visual projection neurons were treated as sources and descending neurons as targets. Only pairs with at least 200 synapses were considered, and every neuron on a path of at most two steps from a source to a target was retained: 52 inputs, 63 outputs and 18 intermediate neurons (133 in total). All 650 connections with at least five synapses between the retained neurons were kept, with weight $W_{ij} = \pm \log(1 + n_{ij})$ for n_{ij} synapses. Signs were assigned from the predicted transmitter of the presynaptic cell [2]: acetylcholine positive (400 connections), GABA (46) and glutamate (196) negative, monoamines (8) positive. The matrix was rescaled to spectral radius 0.9; 62.8% of its connections are excitatory (Fig. 1).

Task. The board is 10×10 and the snake starts with length three. At each step a left turn, a right turn or a straight move is chosen. Food is placed on a uniformly random free cell drawn from a seeded 32-bit xorshift generator, so that a seed fixes the whole game. A game ends on a wall or body collision, or after 200 steps without food. A death is additionally labelled *trapped* if no safe action existed at that moment.

Sensors and network. Fifteen readings are computed in the head frame: the signed forward and lateral offset of the food, its normalised distance and four quadrant flags; three flags for an immediate collision; the inverse distance to the first obstacle along each of the three directions; and two diagonal occupancy flags. Neuron states follow $x \leftarrow x + a\,(-x + g\,W \tanh(x) + I)$ with ten updates per game step, where I is constant within each of eight input groups. Three action scores are read from the mean activity of eight output groups, and the largest is selected. The trainable parameters are the input map, the readout, the leak a and the gain g: 157 values. Training was performed with CMA-ES [3] (population 16, four games per candidate) and stopped after 4,000,000 game steps, about 70 s per run on eight CPU cores. Disjoint seed ranges were used for training, validation and test games, and every configuration was repeated with 10 seeds.

Controls. With identical sensors, interface and budget, W was replaced by a degree-preserving shuffle, by an Erdős–Rényi graph of the same size, by the same graph with excitatory and inhibitory signs permuted, and by $W = 0$, in which case the action is read directly from the driven input groups.

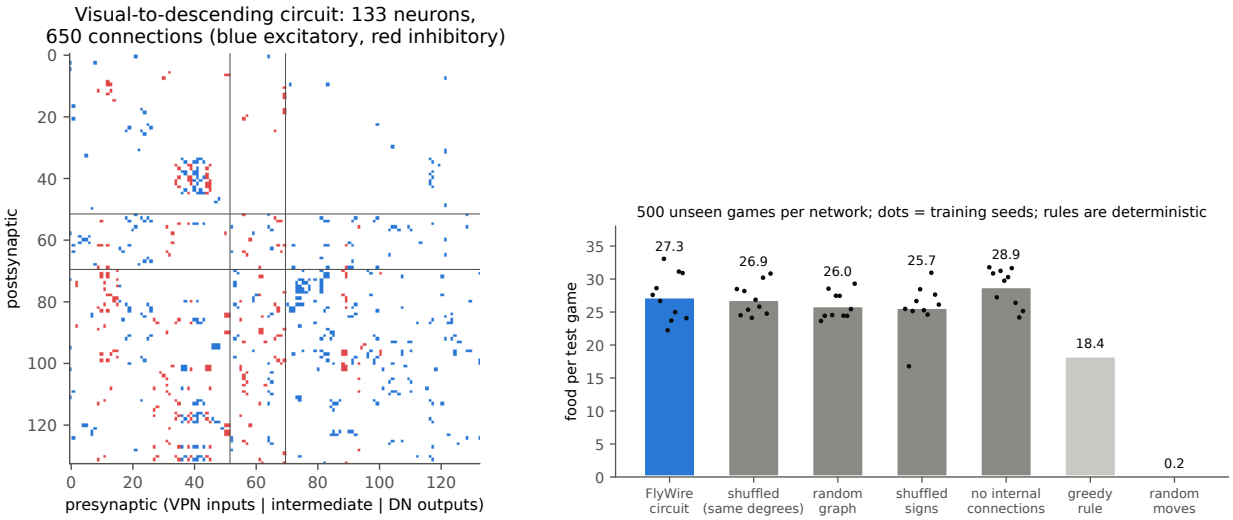


Figure 1: Left: the extracted circuit (inputs, intermediates, outputs; blue excitatory, red inhibitory). Right: food per test game; dots are independent training runs, the two right-hand bars are deterministic reference rules.

3 Results

On average 27.3 ± 3.6 food items per test game were eaten by the trained networks (range 22.2–33.1 across runs; longest snake 53 cells; 18.7 steps per food). A greedy hand-written rule that moves toward the food unless the move is immediately fatal reached 18.4, and uniformly random actions reached 0.2. A flood-fill rule that also keeps the tail reachable, included only as a strong reference, reached 68.8.

Games almost never ended at a wall (0.0%) or in a timeout (0.1%). Instead, 56.5% ended with the snake entering its own body although a safe action was available, and 43.4% ended with no safe action left. The static boundary is thus learned quickly, whereas the growing body is handled poorly (Fig. 2).

Little changed when the measured wiring was replaced: 26.9 ± 2.4 was obtained for the degree-preserving shuffle, 26.0 ± 2.0 for the random graph, 25.7 ± 3.7 for shuffled signs and 28.9 ± 2.9 without recurrence. None of these differed clearly from the connectome (two-sided Mann–Whitney tests over runs: $p = 0.91$, 0.47, 0.62, 0.31). Circuits of 73 and 278 neurons obtained with the same rule scored 27.5 and 26.5, and 25.9 ± 2.6 was obtained when the three obstacle-distance readings were removed. Silencing the 7 most active neurons was worse than silencing the same number at random in only 3 of 10 runs.

4 Discussion

Snake is played at beginner level by a frozen fragment of fly wiring with 157 trained numbers. The comparison with matched graphs shows that, in this setting, the specific connectivity is not what makes it work: most of the task is already carried by the hand-designed sensors and the small trained map, which is why the recurrence-free variant performs equally well. This is a statement about the setup, not about the fly. The task is unrelated to the natural function of these neurons, spiking dynamics are collapsed into a rate variable, transmitter-based signs are predictions rather than measurements [2], and the circuit selection rule (two synapses, strong connections) is one

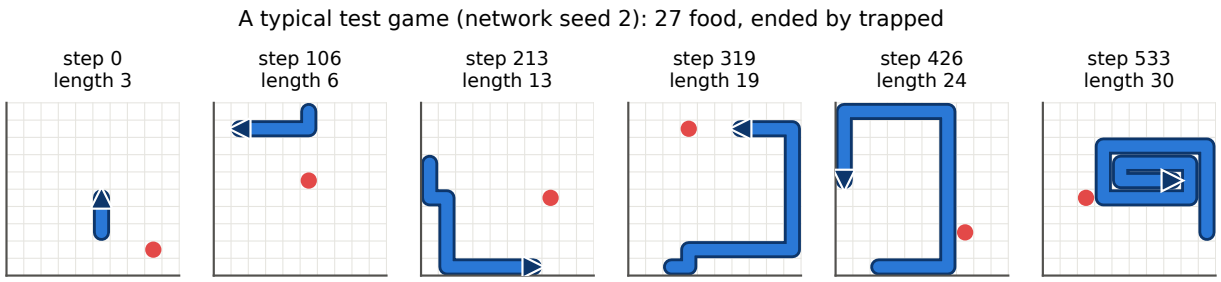


Figure 2: Snapshots of one test game; the triangle marks the head and its heading.

choice among many. The picture might change with a richer task, a longer budget, or a readout constrained to biologically meaningful cell types. The dominant failure mode, running into the snake's own body, suggests that memory of the recent trajectory is the missing ingredient.

5 Conclusion

Connectome-derived wiring can be placed into a closed-loop game agent at laptop scale, and the agent works, but on Snake it is not better than a random graph of the same size.

Data and code availability. FlyWire v783 tables are distributed through the FlyWire Codex [1, 7]. The code, configuration files, per-run logs and the browser demo are available at `https://github.com/burakgulmez/fly-snake`.

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